

E-Commerce Growth & Marketing Analytics

Portfolio Case Study | Historical Public Dataset

Project Overview

This project analyzes historical e-commerce transactions from the publicly available Olist Brazilian E-Commerce dataset. The objective is to understand sales performance, customer purchasing patterns, geographic sales, delivery performance, and customer reviews.

I used Python to clean the data, SQL to answer seven business questions, and Power BI to build an interactive dashboard. The analysis identifies patterns in the recorded transactions and develops evidence-based recommendations for further business investigation.

This is an independent portfolio project using public data, not work completed for a paying client. The findings describe the dataset's historical observation period and should not be interpreted as current marketplace performance or verified business improvements.

1. Business Problem

An e-commerce marketplace generates data across orders, products, customers, payments, deliveries, and customer reviews. When this information is stored in separate tables, it can be difficult to understand overall business performance and identify areas that need attention.

This project addresses seven business questions covering sales trends, product category performance, customer purchasing patterns, geographic sales, delivery performance, customer reviews, and evidence-based business recommendations.

The analysis aims to identify where recorded sales are concentrated, how customers purchase, and how delivery performance is associated with customer ratings.

Project objective: Build an interactive Power BI dashboard and translate the historical transaction data into clear, actionable business insights while distinguishing observed patterns from unverified causes or predicted results.

2. Dataset Description

This project uses the **Olist Brazilian E-Commerce Public Dataset**, which contains historical marketplace transactions primarily from 2016–2018.

The dataset includes nine interconnected tables covering orders, customers, order items, products, sellers, payments, reviews, product category translations, and geolocation.

Data preparation: Python was used to inspect and clean the nine tables before importing them into SQL and Power BI. The cleaned Orders table contains 99,441 records, while the Order Items table contains 112,650 records.

Unit of analysis: Different tables represent different entities. For example, one order can contain multiple products, payments, or review records. The analysis accounts for these relationships to avoid double-counting.

Currency: Monetary values are reported in Brazilian reais (BRL). Delivered product sales are calculated using recorded product prices, excluding freight. They should not be interpreted as profit.

Data limitations: The dataset represents a historical observation period and does not contain all the information needed to calculate profitability, marketing return on investment, or customer lifetime value.

3. Tools & Methodology

Python — Data cleaning: Inspected and cleaned the nine Olist CSV tables, checked data quality, and prepared the files for analysis.

SQL (DuckDB) — Business analysis: Analyzed sales performance, product categories, customer purchasing patterns, geographic sales, delivery performance, and customer reviews. Validated key calculations and prepared evidence-based business recommendations.

Power BI — Data modeling and visualization: Imported the nine cleaned tables, configured seven relationships, created 14 DAX measures, and developed an interactive report with Executive Overview, Sales & Geographic Analysis, and Delivery & Customer Experience pages.

Analysis approach: Defined business questions, prepared the data, calculated and validated metrics, identified patterns, and translated the findings into business insights and recommendations.

4. Key Findings

BRL 13.22M Delivered product sales	96,478 Delivered orders	6.77% Late-delivery rate	3.00% Repeat-customer rate
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Sales performance: Delivered product sales totaled **BRL 13,221,498.11** across **96,478** delivered orders, with average delivered product sales of **BRL 137.04** per delivered order. These figures exclude freight and do not represent profit.

Geographic sales: São Paulo state contributed **38.33%** of delivered product sales. São Paulo, Rio de Janeiro, and Minas Gerais together accounted for **63.38%**.

Customer purchasing: Of **93,358** customers with at least one delivered order, **2,801 (3.00%)** made multiple delivered purchases during the observation period. Repeat customers contributed **5.51%** of delivered product sales.

Delivery performance: The average delivery time was **12.09 days**, and **6.77%** of delivered orders with recorded delivery dates were classified as late.

Customer experience: Late orders received an average review score of **2.27/5**, compared with **4.29/5** for orders classified as not late. This is an observed association, not proof that late delivery caused lower ratings.

Product categories: Health & Beauty generated the highest delivered product sales at **BRL 1,233,131.72**. Office Furniture had an average review score of 3.64/5 across 1,244 reviewed, delivered orders.

5. Business Recommendations

Recommendation 1 — Investigate and reduce late deliveries

Late orders received an average review score of 2.27/5, compared with 4.29/5 for orders classified as not late. Review delivery processes, starting with locations such as Rio de Janeiro, where the recorded late-delivery rate was 12.11%.

Examine seller dispatch times, carrier performance, and delivery routes to identify potential sources of delay. Measure changes using late-delivery rate, average days late, and customer review scores.

Recommendation 2 — Investigate repeat purchasing

Only 3.00% of customers with delivered orders made multiple delivered purchases during the dataset's observation period.

Survey first-time customers about their purchasing experience and examine repeat purchasing by product category. Track the percentage of first-time customers who place a second delivered order within a defined period, such as 90 days.

Recommendation 3 — Investigate office-furniture customer experience

Office Furniture received an average review score of 3.64/5. Even orders classified as not late averaged 3.75/5.

Review customer feedback to identify recurring complaints about product condition, packaging, assembly instructions, or seller service. Track average review scores and low-rating rates after any operational changes.

6. Project Limitations

Historical data: The Olist dataset primarily covers transactions from 2016–2018. The findings describe the recorded observation period and do not represent current marketplace performance.

Public dataset: This is an independent portfolio project, not an analysis performed for a paying client. No business improvements were implemented or measured as part of this project.

Profitability: The dataset does not provide sufficient information about product costs, operating expenses, or marketing expenditure to calculate profit or marketing return on investment.

Customer retention: Purchases outside the dataset's observation period are unavailable. The observed repeat-customer rate should not be interpreted as a lifetime retention rate.

Customer experience: The relationships between delivery performance, product categories, and review scores are observational. They do not prove what caused customers to leave particular ratings.

Data quality: Some delivered orders lack actual delivery dates, and some orders have multiple review records. These issues were accounted for when calculating delivery and review metrics.

7. Project Outcome

This independent portfolio project produced a cleaned, nine-table e-commerce dataset, a SQL analysis addressing seven business questions, and an interactive Power BI report containing 14 DAX measures.

The report brings together sales performance, geographic sales, customer purchasing patterns, delivery performance, and customer reviews.

The analysis identified five key business insights and developed three evidence-based recommendations for further investigation.

Skills demonstrated: Python data cleaning, SQL analysis, relational data modeling, DAX, Power BI dashboard development, data validation, business analysis, and communication of findings to a business audience.

Business impact: The project demonstrates how historical transaction data can support business decision-making. The recommendations were not implemented or tested, so no actual revenue increase, cost reduction, or improvement in customer satisfaction is claimed.

8. Project Links

GitHub Repository: github.com/jackson-david-j/ecommerce-growth-marketing-analytics

The repository includes the Python data-cleaning notebook, SQL analysis notebook, Power BI dashboard previews, and project documentation.